

3/4/2025

FUNC VARIAS VARIABLES; CONTINUIDAD; Derivabilidad; Diferenciabilidad. Superficies

⇒ CONJUNTOS DE NIVEL

* Ejercicio 1 → Det dom f, graf dom f analizar tipode conjunto, Conj nivel f y graf N_k

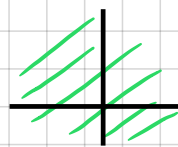
a) $f(x,y) = 3(1 - \frac{x}{2} - \frac{y}{2})$

• Dom f = $\{(x,y) \in \mathbb{R}^2\} \rightarrow$ conjunto abierto no acotado (todo $\mathbb{R}^2$)

• Conjunto de nivel k

$$N_k = \{(x,y) \in \mathbb{R}^2 / 3(1 - \frac{x+y}{2}) = k, k \in \mathbb{R}\}$$

$$3(1 - \frac{x+y}{2}) = k \Rightarrow 1 - \frac{k}{3} = \frac{x+y}{2} \Rightarrow 2 - \frac{2k}{3} = x+y \Rightarrow \text{rectas}$$



b) $f(x,y) = \sqrt{y-x}$



• Hallar dom $y-x \geq 0 \Rightarrow y \geq x$ Dom f = $\{(x,y) \in \mathbb{R}^2 / y \geq x\} \rightarrow$ cerrado, no acotado

$\nexists B(r,A)$ con $r \in \mathbb{R}$

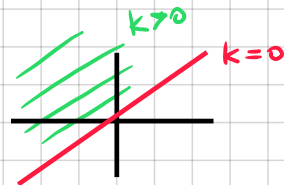
• Hallar conj Nivel N_k

$$N_k = \{(x,y) \in D / \sqrt{y-x} = k\}$$

$$\sqrt{y-x} = k \rightarrow k=0 \Rightarrow \sqrt{y-x} = 0 \Rightarrow \boxed{y=x} \text{ recta identidad}$$

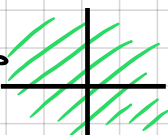
$$\rightarrow k > 0 \Rightarrow \sqrt{y-x} = k \Rightarrow \boxed{y = k^2 + x} \text{ rectas pendiente } m=1 \text{ ord origen } b=k^2$$

$$\rightarrow k < 0 \Rightarrow \nexists \sqrt{y-x} = k \quad \phi$$



c) $f(x,y) = 25 - x^2$

conj abierto no acotado



• Dom (f) = $\{(x,y) \in \mathbb{R}^2\}$

• Conj Nivel $k = \{(x,y) \in \mathbb{R}^2 / 25 - x^2 = k\}$

$$25 - x^2 = k \Rightarrow x^2 = \frac{25-k}{25}$$

$$\boxed{25-k \geq 0} \\ \boxed{k \leq 25}$$

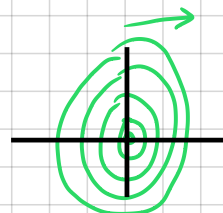
$$N_k = \{(x,y) \in \mathbb{R}^2 / 25 - x^2 = k, k \leq 25\}$$

Si $k=25 \rightarrow x=0$

Si $k < 25 \rightarrow |x| = \sqrt{25-k} \rightarrow$ rectas (x,y) en $\mathbb{R}^2$ verticales

$k > 25 \rightarrow \phi$

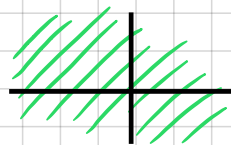
$k=25$



d) $f(x,y) = 9x^2 + 4y^2$

• Dom (f) = $\{(x,y) \in \mathbb{R}^2\}$

• Conjunto de nivel $N_k = \{(x,y) \in \mathbb{R}^2 / 9x^2 + 4y^2 = k\}$



conj abierto no acotado

$$9x^2 + 4y^2 = k$$

$$\frac{9x^2}{k} + \frac{4y^2}{k} = 1 \Rightarrow \frac{x^2}{(\frac{k}{9})} + \frac{y^2}{(\frac{k}{4})} = 1 \rightarrow \text{elipses}$$

$$\left. \begin{aligned} a &= \sqrt{\frac{k}{9}} \\ b &= \sqrt{\frac{k}{4}} \end{aligned} \right\} \rightarrow \boxed{k \geq 0}$$

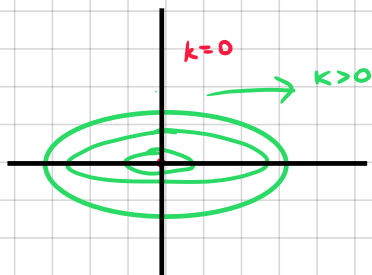
$\boxed{k > 0}$

Si $k > 0 \rightarrow$ elipse
Si $k = 0 \rightarrow \phi$
Si $k < 0 \rightarrow \phi$

e) $f(x,y) = \sqrt{x^2+2y^2}$

• Dom(f) = $\{(x,y) \in \mathbb{R}^2\}$

$x^2+2y^2 \geq 0$
siempre positivo



• Conjunto Nivel $N_k = \{(x,y) \in \mathbb{R}^2 / \sqrt{x^2+2y^2} = k\}$

$\sqrt{x^2+2y^2} = k \rightarrow |x^2+2y^2| = k^2 \rightarrow x^2+2y^2 = k^2$
 $\rightarrow \frac{x^2}{(k^2)} + \frac{y^2}{(\frac{k^2}{2})} = 1$, donde $k > 0$
elipse

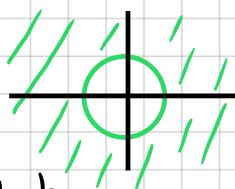
$\left\{ \begin{array}{l} \text{si } k > 0 \rightarrow \frac{x^2}{k^2} + \frac{y^2}{\frac{k^2}{2}} = 1 \rightarrow \text{elipses} \\ \text{si } k = 0 \rightarrow \sqrt{x^2+2y^2} = k^2 \rightarrow (0,0) \\ \text{si } k < 0 \rightarrow \emptyset \end{array} \right.$

f) $f(x,y) = (x^2+y^2-16)^{-\frac{1}{2}} = \frac{1}{\sqrt{x^2+y^2-16}}$

Dom(f) $\rightarrow \{(x,y) \in \mathbb{R}^2 / x^2+y^2 > 16\}$

$x^2+y^2-16 > 0$

$x^2+y^2 > 16$



conjunto abierto, no acotado

• $N_k = \{(x,y) \in \mathbb{D} / \frac{1}{\sqrt{x^2+y^2-16}} = k\}$

$\frac{1}{\sqrt{x^2+y^2-16}} = k$

$k \sqrt{x^2+y^2-16} = 1 \rightarrow x^2+y^2-16 = \frac{1}{k^2}$
 $\downarrow$
 $\neq 0$ En Dom(f) $\rightarrow x^2+y^2 = \frac{1}{k^2} + 16$
 $\searrow$
 > 0

Hallar k para $\frac{1}{k^2} + 16 \geq 0$

$\frac{1}{k^2} + 16 > 0 \quad -16k < 1$

$\frac{1}{k^2} > -16 \quad k > \frac{1}{-16}$

• Niveles disponibles (k)

si $k = 0 \rightarrow \emptyset$

si $k > \frac{1}{16} \wedge k \neq 0 \rightarrow \text{circunferencias}$

si $k \leq \frac{1}{16} \rightarrow \emptyset$

g) $f(x,y) = e^{-x^2-y^2} = e^{-(x^2+y^2)} = \frac{1}{e^{x^2+y^2}}$

Dom(f) = $\{(x,y) \in \mathbb{R}^2\}$

$e^{x^2+y^2} \geq 0$
 ≥ 0

función exponencial
nunca negativo

• Hallar conjunto Nivel : $\{(x,y) \in \mathbb{R}^2 / e^{-x^2-y^2} = k\}$

$\frac{1}{e^{x^2+y^2}} = k \rightarrow e^{x^2+y^2} = \frac{1}{k}$

$x^2+y^2 = \ln\left(\frac{1}{k}\right)$, $k \neq 0 \wedge \frac{1}{k} > 0$
Circunferencias $k > 0$

$\left\{ \begin{array}{l} \text{si } k = 1 \rightarrow x^2+y^2 = 0 \rightarrow (0,0) \\ \text{si } k > 0 \wedge k \neq 1 \rightarrow x^2+y^2 = \ln\left(\frac{1}{k}\right) \rightarrow \text{circunferencias} \\ \text{si } k < 0 \rightarrow \emptyset \end{array} \right.$
radio $\sqrt{\ln\left(\frac{1}{k}\right)}$

h) $f(x, y, z) = \ln(4 - x - y)$

• Hallar dominio

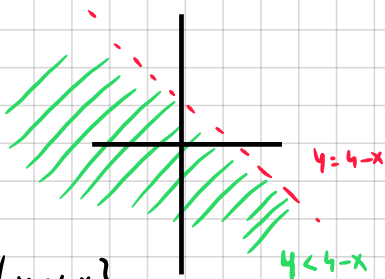
$$4 - x - y > 0$$

$$4 - x > y$$

$$\text{Dom}(f) = \{(x, y, z) \in \mathbb{R}^3 \mid y < 4 - x\}$$

→ conj abierto → no incluye frontera $4 - x = y$

→ no acotado



• Hallar conjunto nivel $N_k = \{(x, y, z) \in \mathbb{R}^3 \mid \ln(4 - x - y) = k\}$

$$\ln(4 - x - y) = k$$

$$4 - x - y = e^k$$

$$4 - e^k - x = y$$

plano en $\mathbb{R}^3$



i) $f(x, y, z) = x + y + 2z$

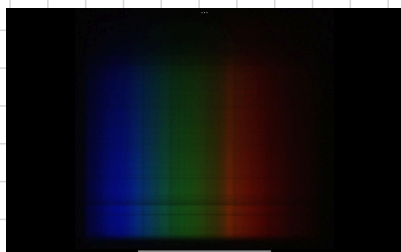
$$\text{Dom}(f) = \mathbb{R}^3$$

$$N_k = \{(x, y, z) \in \mathbb{R}^3 \mid \underbrace{x + y + 2z = k}_{\text{plano en } \mathbb{R}^3}\}$$

j) $f(x, y, z) = e^{x^2 + y^2 - z^2}$

$$\text{Dom}(f) = \{(x, y, z) \in \mathbb{R}^3 \mid \text{función exponencial}\}$$

analiza en $\mathbb{R}^3$



$$N_k = \{(x, y, z) \in \mathbb{R}^3 \mid e^{x^2 + y^2 - z^2} = k\}$$

$$e^{x^2 + y^2 - z^2} = k$$

$$x^2 + y^2 - z^2 = \ln(k)$$

→ $\in \mathbb{R}$

$$\text{si } k \leq 0 \rightarrow N_k = \emptyset$$

$$\frac{x^2 + y^2 - \ln(k)}{-2} = z \Rightarrow z = \underbrace{\frac{-x^2}{2} - \frac{y^2}{2} + \ln(k)}_{(?)}$$

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* Ejercicio 2 = Graf, si quedan def, No y N_4

a) $f(x, y) = \begin{cases} x + y & \text{si } x \geq -2 \\ 0 & \text{si } x < -2 \end{cases}$

• Hallar Dominio $\text{Dom}(f) = \{(x, y) \in \mathbb{R}^2\}$

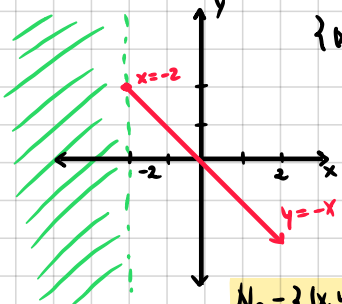
• Hallar $N_0 \rightarrow \{(x, y) \in D \mid f(x, y) = 0\}$

$$\{(x, y) \in D \mid x < -2\}$$

¿Qué pasa si $x \geq -2$ y $f(x, y) = 0$

$$x + y = 0 \Rightarrow y = -x$$

$$N_0 = \{(x, y) \in \mathbb{R}^2 \mid x < -2\} \cup \{(x, y) \in D \mid y = -x, x \geq -2\}$$



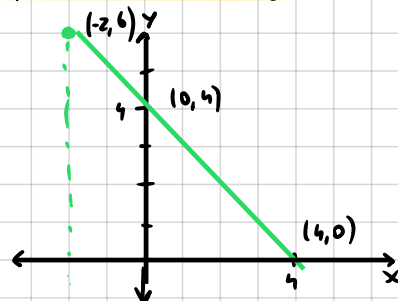
• Hallar $N_4 = \{(x, y) \in D \mid f(x, y) = 4\}$

si $x \geq -2$

$$x + y = 4 \Rightarrow y = 4 - x$$

recta

$$N_4 = \{(x, y) \in D \mid y = 4 - x \wedge x \geq -2\}$$



* ⑥ $f(x,y) = \sin(y-x) \rightarrow$ función periódica $\rightarrow f(x,y) \in [0,1]$

• Hallar dominio $\text{Dom } f = \{(x,y) \in \mathbb{R}^2\}$

• Gráfico No

• Hallar conjunto Nivel 0

$$f(x,y) = \sin(y-x) = 0$$

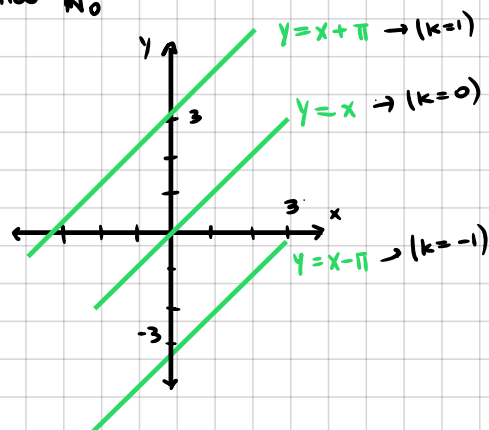
función periódica

$$y-x = \arcsen(0) = \pi k, k \in \mathbb{Z}$$

$$y = x + \pi k \Rightarrow \text{rectas en } \mathbb{R}^2$$

$$N_4 = \emptyset$$

• Hallar $N_4 \rightarrow \nexists f(x,y) = 4 \rightarrow f(x,y)$ oscila entre $[0,1]$



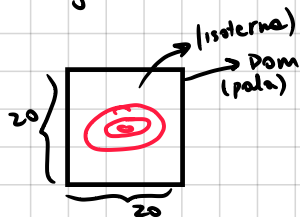
* $D = \{(x,y) \in \mathbb{R}^2 : x \in [-10,10], y \in [-10,10]\} = [-10,10] \times [-10,10] \rightarrow$ placa metálica

$$T(x,y) = 64 - 4x^2 - 8y^2 \rightarrow \text{temperatura de punto } (x,y) \in D$$

Dibujar "isotermas" $\rightarrow$ Conj de puntos con misma temperatura k

• Figura de análisis

• Hallar isotermos de Temperatura k
• Hallar conjunto de nivel N_k (en general)



$$T(x,y) = 64 - 4x^2 - 8y^2 = k$$

$$64 - k = 4x^2 + 8y^2 \Rightarrow \text{elipses}$$

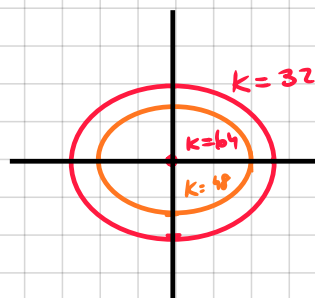
$$1 = \frac{4x^2 + 8y^2}{64 - k} \Rightarrow 1 = \frac{x^2}{\left(\frac{64-k}{4}\right)} + \frac{y^2}{\left(\frac{64-k}{8}\right)} \quad \begin{matrix} 64-k > 0 \\ |k| < 64 \end{matrix}$$

• Gráfico

$$k = 32 \Rightarrow 1 = \frac{x^2}{8} + \frac{y^2}{4} \quad \begin{matrix} a = \sqrt{8} \approx 2,82 \\ b = 2 \end{matrix}$$

$$k = 48 \Rightarrow 1 = \frac{x^2}{4} + \frac{y^2}{2} \quad \begin{matrix} a = 2 \\ b = \sqrt{2} \approx 1,41 \end{matrix}$$

$$k = 64 \Rightarrow 0 = 4x^2 + 8y^2 \rightarrow (0,0)$$



$$k = 16 \Rightarrow 1 = \frac{x^2}{12} + \frac{y^2}{6}$$

$$k = 8 \Rightarrow 1 = \frac{x^2}{14} + \frac{y^2}{7}$$

* Ejercicio 4 $U(x,y)$ esp potencial electrostático $(x,y) \in D \subset \mathbb{R}^2$

$$U(x,y) = \frac{k}{\sqrt{x^2 + y^2}} \quad (x,y) \in \mathbb{R}^2 - \{0\} \quad \text{siendo } k \in \mathbb{R} > 0$$

• Hallar líneas equipotenciales (E)

$$U(x,y) = \frac{k}{\sqrt{x^2 + y^2}} = E \quad \begin{matrix} > 0 \text{ y } (x,y) \in \mathbb{R}^2 \end{matrix}$$

$$\left(\frac{k}{E}\right)^2 = x^2 + y^2$$

$\rightarrow$ circunferencias en $\mathbb{R}^2$
de radio $r = \frac{k}{E}$, $y E > 0$

no existe equipotencial 0

LÍMITES Y CONTINUIDAD DE CAMPOS

* Ejercicios = Determinar existencia del lím, fundamentar rta

a) $\lim_{(x,y) \rightarrow (1,1)} xy - y^2 = 0 \Rightarrow \exists \lim_{(x,y) \rightarrow (1,1)} (xy - y^2) = 0$ (existe lím y es 0)

b) $\lim_{(x,y) \rightarrow (0,4)} xy^{-\frac{1}{2}} = \lim_{(x,y) \rightarrow (0,4)} \frac{1}{\sqrt{xy}} = \infty$ no existe lím

c) $\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} = 0 \Rightarrow \exists \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} = 0$ existe lím y es 0

d) $\lim_{(x,y) \rightarrow (0,0)} \frac{x}{x+y} \Rightarrow \text{ind } \frac{0}{0} \rightarrow \text{probar acercamiento } (0,0) \text{ por curvas}$

• familia de rectas $(0,0) \rightarrow y = mx$

$f(x, mx) = \frac{x}{x+mx} = \frac{x}{x(1+m)}$

$\Rightarrow \lim_{x \rightarrow 0} \frac{x}{x(1+m)} = \lim_{x \rightarrow 0} \frac{1}{1+m} = \frac{1}{1+m}$

si $m=1 \rightarrow \lim_{x \rightarrow 0} f(x, xm) = \frac{1}{2}$
si $m=2 \rightarrow \lim_{x \rightarrow 0} f(x, mx) = \frac{1}{3}$

$\hookrightarrow$ si límite existe es único $\rightarrow \nexists \lim_{(x,y) \rightarrow (0,0)} \frac{x}{x+y}$

e) $\lim_{(x,y) \rightarrow (0,2)} \frac{x^2(y-2)^2}{x^2 + (y-2)^2} = \text{ind } \frac{0}{0} \Rightarrow \lim_{(x,y) \rightarrow (0,2)} \frac{x^2(y-2)^2}{x^2 + (y-2)^2} = 0$

$\Rightarrow$ probar acotado

$x^2 \leq \frac{x^2 + (y-2)^2}{\geq 0} \rightarrow \frac{x^2}{x^2 + (y-2)^2} \leq 1 \rightarrow \underbrace{\left| \frac{x^2}{x^2 + (y-2)^2} \right|}_{\geq 0} \leq \frac{1}{1} \forall (x,y) \in \mathbb{R}^2$

f) $\lim_{x \rightarrow 0} \left(\frac{x}{|x|}, \sqrt{x} \right) f_1 = \begin{cases} \frac{x}{x} & \text{si } x > 0 \\ -\frac{x}{x} & \text{si } x < 0 \end{cases}$

• En f_1 , $\lim_{x \rightarrow 0} f_1(x) = \begin{cases} \lim_{x \rightarrow 0^+} \frac{x}{x} = 1 \\ \lim_{x \rightarrow 0^-} -\frac{x}{x} = -1 \end{cases} \rightarrow \text{como } \nexists \lim_{x \rightarrow 0} f_1, \text{ entonces } \nexists \lim_{x \rightarrow 0} f(x)$

• técnicas resolución límites

- $\hookrightarrow$ Probar que no existen $\rightarrow$ probar por curvas
- $\hookrightarrow$ Probar 0 $\rightarrow$ probar acotado
- $\hookrightarrow$ Cambio de variable
- $\hookrightarrow$ simplificar
- $\hookrightarrow$ parametrización

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$$g) \lim_{(x,y) \rightarrow (0,0)} \left(\underbrace{y \cos(x^{-1})}_{f_1}, \underbrace{\frac{\sin(3x)}{2x}}_{f_2}, \underbrace{x-y}_{f_3} \right) = \lim_{(x,y) \rightarrow (0,0)} (f_1(x,y), f_2(x,y), f_3(x,y))$$

$$\begin{aligned} \bullet \lim_{(x,y) \rightarrow (0,0)} y \cos(x^{-1}) &= 0 \quad \text{acotado en } [0,1] \\ \bullet \lim_{(x,y) \rightarrow (0,0)} x-y &= 0 \\ \bullet \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(3x)}{2x} &= \lim_{(x,y) \rightarrow (0,0)} \frac{\cos(3x)}{2} = \frac{1}{2} \end{aligned} \quad \left\{ \begin{aligned} \lim_{(x,y) \rightarrow (0,0)} (y \cos(x^{-1}), \frac{\sin(3x)}{2x}, x-y) &= (0, \frac{1}{2}, 0) \end{aligned} \right.$$

L'Hop

$$h) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2-y^2)}{x-y} = \text{Ind}$$

• Probar con recta $y=mx$

$$\lim_{x \rightarrow 0} \frac{\sin(x^2-m^2x^2)}{x-mx} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{\cos(x^2-m^2x^2) (2x-2mx)}{1-m} = 0$$

• Probar parabolas $y=kx^2$

$$\lim_{x \rightarrow 0} \frac{\sin(x^2-k^2x^4)}{x-kx^2} = \lim_{x \rightarrow 0} \frac{\cos(x^2-k^2x^4) (2x-4x^3k^2)}{1-k} = 0$$

$$\frac{\sin(x^2-y^2)}{x-y} = \frac{\sin((x+y)(x-y))}{(x-y)} = \frac{\sin(x^2-y^2)(x+y)}{(x-y)(x+y)}$$

$$\left(\frac{\sin(x^2-y^2)}{x^2-y^2} \right) x + \left(\frac{\sin(x^2-y^2)}{x^2-y^2} \right) y$$

$$\lim_{x,y \rightarrow 0,0} \left[\frac{\sin(x^2-y^2)}{x^2-y^2} x + \frac{\sin(x^2-y^2)}{x^2-y^2} y \right] = 0$$

A chequar

$$i) \lim_{(x,y) \rightarrow (0,0)} \frac{e^{x^2+y^2}-1}{x^2+y^2} = \text{Ind } \frac{0}{0}$$

$$\boxed{\begin{aligned} M &= x^2+y^2 \\ \lim_{x,y \rightarrow 0,0} M &= 0 \end{aligned}}$$

$$\lim_{M \rightarrow 0} \frac{e^M-1}{M} = \lim_{M \rightarrow 0} \frac{e^M}{1} = 1$$

$$j) \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^2} = \lim_{(x,y) \rightarrow (0,0)} x \cdot \frac{y^2}{x^2+y^2} \quad \text{acotado en } B^*(0,0,\epsilon)$$

$$k) \lim_{(x,y) \rightarrow (0,0)} \frac{y^5}{x^2+y^4} = \text{Ind } \frac{0}{0} \Rightarrow \lim_{(x,y) \rightarrow (0,0)} y \cdot \frac{y^4}{x^2+y^4} = 0 \quad \text{acotado } B^*(0,0,\epsilon)$$

$$y^4 \leq x^2+y^2$$

$$\left| \frac{y^4}{x^2+y^2} \right| \leq 1 \rightarrow \left| \frac{y^5}{x^2+y^4} \right| \leq |y|$$

• analizar curva $y=mx$

$$\lim_{x \rightarrow 0} \frac{(mx)^5}{x^2+(mx)^4} = \lim_{x \rightarrow 0} \frac{x^5 m^5}{x^2(1+m^4x^2)} = \lim_{x \rightarrow 0} \frac{x^3 m^5}{(1+m^4x^2)} = \lim_{x \rightarrow 0} \frac{3x^2 m^5}{2m^4 x} = \lim_{x \rightarrow 0} \frac{3xm^5}{2m^4} = 0$$

• analizar $x=ky^2$

$$\lim_{y \rightarrow 0} \frac{y^5}{k^2 y^4 + y^4} = \lim_{y \rightarrow 0} \frac{y^5}{y^4(k^2+1)} = \lim_{y \rightarrow 0} \frac{y}{k^2+1} = 0$$

① $\lim_{(x,y) \rightarrow (0,0)} (x^2+y^2) \ln(x^2+y^2) = \lim_{u \rightarrow 0} \underbrace{u}_{\rightarrow 0} \underbrace{\ln(u)}_{\rightarrow -\infty} = \text{Ind } 0 \cdot \infty$ $\lim_{u \rightarrow 0} u \ln u = \lim_{u \rightarrow 0} \frac{\ln(u)}{(\frac{1}{u})} = \lim_{u \rightarrow 0} \frac{(\frac{1}{u})}{-\frac{1}{u^2}} = \lim_{u \rightarrow 0} -\frac{1}{u} \frac{u^2}{1} = 0$

$\boxed{\begin{matrix} u = x^2+y^2 \\ \lim_{x,y \rightarrow 0,0} u = 0 \end{matrix}}$

$\lim_{u \rightarrow 0} u \ln(u) = \lim_{u \rightarrow 0} \ln(u^u) = \lim_{u \rightarrow 0} \ln \left[\left(1 + \frac{u^u - 1}{u^u}\right)^{u \cdot \frac{1}{u^u - 1}} \right] = 1$

② $\lim_{(x,y) \rightarrow (1,-1)} \frac{x^3+y^3}{x+y} \xrightarrow{0} \Rightarrow \text{división polinómica}$

$$\begin{array}{r} x^3+y^3 \quad | \quad x+y \\ -x^3+y^2-xy \\ \hline y^3-y^2x \\ -y^3+y^2x \\ \hline -yx^2-y^2x \\ -yx^2-y^2x \\ \hline 0 \end{array}$$

$= \lim_{(x,y) \rightarrow (1,-1)} \frac{x^2+y^2-xy}{1} = 3$

③ $\lim_{(x,y) \rightarrow (1,1)} \frac{2y^2-x^2-xy}{y-x} \xrightarrow{0} = \lim_{(x,y) \rightarrow (1,1)} \frac{y^2+y^2-x^2-xy}{(y-x)} = \lim_{x,y \rightarrow (1,1)} \frac{(y-x)(y+x) + y(y-x)}{(y-x)} = \lim_{x,y \rightarrow (1,1)} y+x+y = 3$

* Ejercicio 6